



The shifting happiness return to higher education over 50 years: causal mediation analysis of college completion, economic benefits, and happiness in the USA

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Abstract

Evidence shows higher education (HE) leads to such socio-economic outcomes as better jobs, income, and happiness. While this implies HE demonstrates multifaceted values incorporating economic benefits and non-pecuniary public good, their causal relationship and longitudinal trends remain empirically elusive. Using data from the General Social Survey over 50 years in the USA (1973–2022, $N=35,483$), we conduct causal mediation analysis to examine the effect of college completion on happiness, including indirect pathways through occupational attainment and income. The results reveal that the overall happiness return to HE has increased over 45 years, followed by a nosedive in 2021–2022 during the COVID-19 pandemic. In the mid-1970s, HE contributed to happiness primarily via occupations, whereas the direct HE-happiness linkage conditional on economic rewards showed a negative sign. However, the relative impact of jobs subsided while the mediating role of income net of occupations increased. Meanwhile, the happiness return to postgraduate degrees has been consistently mediated by occupations. Notably, these premiums for HE are formed in tandem with declining happiness levels among less-educated individuals. We thus argue (1) HE may contribute to relatively better non-pecuniary outcomes like subjective well-being, underpinned by enhanced economic benefits, through different mechanisms for undergraduates and postgraduates; (2) this HE effect emerges alongside marginalization of the less educated; and (3) HE policy/practice should consider both potentially positive and adverse effects on college completers and non-completers to fulfill its public good role. Given that macro-socioeconomic conditions may affect these dynamics, we also call for further comparative studies in this field.

Keywords Higher education · Happiness · Income · Occupation · Public good · Causal mediation analysis

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Introduction

Scholars argue higher education contributes to socio-economic rewards, such as higher income, stable jobs, better health, union formation, and broader social connections (Hout, 2012; OECD, 2007). Recent research (e.g., Araki, 2022; Cuñado & de Gracia, 2012; Helliwell et al., 2024) further detects the positive link between higher education and subjective well-being. Alongside these individual-level mechanisms, societal-level outcomes of higher education have also been extensively explored, ranging from strengthened democracy and integration to equity, social mobility, and peace (Ashwin, 2022). These studies suggest that higher education plays a crucial role in promoting both private benefits and public good (Huang et al., 2022; Marginson, 2007, 2011, 2018; Mihut, 2024; Tian et al., 2025).

Among a range of potential outcomes of higher education, happiness is gaining attention as a form of public good, particularly in relation to human development (Mihut, 2024). From a psychological perspective, higher education may foster a sense of personal control, enabling individuals to cope more effectively with stress and adversity (Chipperfield et al., 2017; Ross & Mirowsky, 2013). College graduates also tend to develop greater patience, a reduced focus on immediate gains, and a stronger future orientation (Oreopoulos & Salvanes, 2011), all of which could support sustained well-being throughout the lifespan. These arguments indicate the nature of higher education as a public good, which transcends its narrowly defined values as a production of labor force and a stepping stone to status attainment (Marginson, 2016b).

Nonetheless, evidence also reveals that the significant contribution of higher education to happiness may disappear, or even become detrimental, once accounting for the mediating effect of pecuniary rewards (Araki, 2022; Clark & Oswald, 1996; Layard, 2011). That is, higher education may lead to better subjective well-being only when the educational attainment brings higher economic standing, which would promote their happiness. Should this be the case, one may argue higher education demonstrates its instrumental utility more substantially than its intrinsic value, such that the direct effect of higher education on happiness is not necessarily guaranteed. Put differently, psychological returns to higher education may be dependent on its economic benefits.

Despite such a nuanced understanding of linkages between higher education and happiness, two fundamental aspects remain empirically elusive: their longitudinal trend and causal relationships. Given that the relative value of tertiary degrees as a positional good changes in tandem with societal-level educational expansion and industrial development (Araki, 2020; Di Stasio et al., 2016; Horowitz, 2018; Triventi et al., 2016; Van de Werfhorst, 2009), it is plausible that the relationship between higher education and multidimensional outcomes shifts dynamically in the process of transitioning to high participation systems (Araki, 2023; Cantwell et al., 2018; Marginson, 2016a, 2016c). Evidence indeed shows that the higher education-happiness association becomes weaker in societies with a larger proportion of tertiary graduates among the population (Araki, 2022). However, while prior studies have delineated a structure at a certain point in time, we know little about how it has evolved over time. In addition, primarily because of data availability, research in this vein often relies on traditional regression models to examine a sign for higher education as a predictor of happiness, including its elasticity when economic benefits are taken into account. Although this approach is useful to explain associations between the variables concerned, one cannot necessarily interpret them as causal structures.

Against this background, the current study leverages causal mediation analysis based on the counterfactual framework to elucidate the effect of college completion on happiness, with particular attention to potential mediation by economic returns and their longitudinal variation. In the next section, we review the relevant literature to formulate hypotheses. Data and methods are then described, followed by analysis results and discussion. As detailed below, our analysis uses nationally representative data from the General Social Survey in the USA over 50 years, illuminating both pecuniary and non-pecuniary values of higher education.

Higher education, economic benefits, and happiness: public good perspective

Scholars have long documented that better-educated individuals are likely to enjoy greater job prospects and higher income (Goldin & Katz, 2008; Hout, 2012). Extending this economic viewpoint, recent research pays close attention to public good outcomes of higher education (Brewis & Marginson, 2025; Chankseliani et al., 2021; Marginson, 2025; McMahon, 2009). For example, Marginson (2011) argues that universities should be seen not merely as providers of workforce training or sources of macroeconomic output but as drivers of democratization and human development in a holistic sense. Marginson (2025) further specifies public good as non-pecuniary benefits for individuals and collective contributions to society. In line with this perspective, a growing body of studies has explored multidimensional functions of higher education, encompassing knowledge diffusion, human mobility, cultural development, and social cohesion (Hazelkorn & Gibson, 2019; Tian et al., 2025; Yang & Chen, 2025). Notably, Mihut (2024) examined happiness as an essential element of human development, hence public good that could be fostered by higher education. This view, treating happiness as important enrichment, aligns with Aristotle's (1968) definition of happiness as the "ultimate goal of human life" and Sen's (1999) capability approach highlighting individuals' freedom and well-being as a primary goal of development.

Indeed, over the past decades, the association between higher education and happiness has been extensively investigated. While the evidence is divided concerning whether education directly heightens happiness, there is a near consensus that college education may lead to better subjective well-being at least indirectly via improved economic status (Araki, 2022; Chen, 2012; Cuñado & de Gracia, 2012; Hu, 2015; Nikolaev, 2018; Ruii & Ruii, 2019). Specifically, one typical mediator operating as both the product of higher education and the determinant of happiness is occupational attainment. College graduates are more likely to find jobs with greater autonomy, intellectual engagement, and higher occupational prestige (Acemoglu & Autor, 2011; Autor, 2015; Oreopoulos & Salvanes, 2011; Solomon et al., 2022). These work arrangements predict higher job satisfaction (Benz & Frey, 2008; Clark, 2010), which is also closely linked to overall happiness (Bowling et al., 2010; Heller et al., 2002; Rain et al., 1991). Nonetheless, it is noteworthy that the positive effect of high-level educational attainment on job satisfaction can be cancelled when educated individuals are exposed to increased work hours and task pressure/intensity along with enhanced resources and autonomy (Solomon et al., 2022).

Another economic aspect that may mediate the association between higher education and happiness is monetary rewards. On the one hand, numerous studies have consistently shown that higher levels of education lead to increased income (e.g., Angrist et al., 2021; Card, 2001; Goldin & Katz, 2008; Sewell & Hauser, 1975). On the other

hand, higher income is considered to bring a better psychological state (Deaton, 2008; Diener et al., 2010; Killingsworth, 2021). While some studies highlight the existence of a saturation point in this relationship, beyond which further increases in monetary assets do not improve happiness (Jebb et al., 2018; Kahneman & Deaton, 2010), recent evidence shows that the saturation is evident primarily among the least happy individuals, whereas happiness steadily increases along with income among happier individuals and even accelerates within the happiest group (Killingsworth et al., 2023).

Apart from the indirect effect of higher education on happiness through improved economic conditions, higher education may exert a direct influence. This is plausible when the attainment of higher education per se elevates positive psychological state through the intrinsic process of learning, fostering a sense of autonomy and choice (Brighouse, 2006; Novarese & Rizzello, 2005) and building up self-esteem and self-confidence (Cuñado & de Gracia, 2012).

Meanwhile, research also shows that the relationship between educational attainment and happiness levels is null, or even negative, when accounting for the function of economic assets. This may happen when highly educated individuals possess higher expectations for economic returns, which are eventually not fulfilled, resulting in dissatisfaction with their life (Clark & Oswald, 1996). Another explanation is that individuals with different education and income levels may diverge in their tendency to report happiness levels using focal values with rounding behaviors, thus incurring bias into the observed associations (Barrington-Leigh, 2024).

As such, the nuanced association among higher education, economic benefits, and happiness has been progressively explained. However, two aspects have been understudied in this field. First, from a methodological perspective, the typical approach involves conducting regressions with education and happiness measures as the predictor and outcome variables, respectively, followed by an extended model that incorporates economic standing as an additional predictor. Scholars often analyze how the education effect on happiness changes in terms of magnitude and significance before and after including economic status in the model. While this approach provides valuable insights into the variation in the education-happiness linkage when economic contributions are considered, it does not necessarily offer robust evidence regarding the causal path from higher education to happiness through labor market outcomes. This is not merely a technical concern but a crucial empirical gap to be addressed for a better understanding of the nature of higher education as a public good. Therefore, we first test the following hypothesis using causal mediation analysis:

Hypothesis 1: Higher education contributes to happiness, both directly and indirectly through improved economic standing.

Second, most prior studies examining the mediating role of economic assets have focused on a single time point. Little is known, however, about the longitudinal trends in the causal structure linking higher education and happiness. Given that the relative value of college completion has shifted in the past decades in tandem with educational proliferation toward high participation systems (Araki, 2023; Cantwell et al., 2018; Horowitz, 2018; Marginson, 2016c), it is plausible that the dynamics involving higher education, economic outcomes, and happiness may have also evolved over time. In particular, evidence suggests that the economic premium associated with tertiary degrees, viewed as a positional good, has declined as these credentials become less scarce due to higher education expansion at the societal level (Araki & Kariya, 2022; Brown, 2001; Brown, 2003; Mok & Jiang, 2018; Ortiz & Rodriguez-Menés, 2016; Tholen, 2017).

Meanwhile, people's awareness of non-pecuniary aspects of life has increased over time. A growing number of studies have documented how fulfilling psychological needs and seeking a sense of meaning can act as powerful motivators that go beyond mere economic incentives (Diener & Seligman, 2004). Moreover, happiness studies underline that non-monetary factors like interpersonal networks play an important role in explaining the education-happiness link (Chen, 2012). Overall, considering the potentially declining economic value of higher education and the rising significance of non-pecuniary factors in shaping happiness, we hypothesize the declining happiness return to higher education as follows:

Hypothesis 2-1: The happiness return to higher education, especially the indirect path through economic benefits, has decreased over the past decades.

On the contrary, some evidence also suggests that the economic returns to college education have actually increased over time in the USA, coinciding with a growing demand for highly educated workers in the labor market (Acemoglu & Autor, 2011; Autor, 2014; Goldin & Katz, 2008; Lemieux, 2008). Furthermore, research on inequality consistently reveals that growing economic disparities are significantly associated with differences in psychological well-being (Cheung & Lucas, 2016; Delhey & Dragolov, 2014; Oishi et al., 2011). If the happiness gap across economic strata remains sizable, coupled with the rising return to college completion, the indirect effect of higher education on happiness via labor market outcomes may have rather strengthened while penalizing the less educated over time.

Hypothesis 2-2: The happiness return to higher education, especially the indirect path through economic benefits, has increased over the past decades.

Herein, as the prevalence of postgraduate programs increases in the higher education sector (Britton et al., 2020; Jung, 2023; Wakeling & Laurison, 2017), one should pay close attention to the potential difference in the structure across degree levels. Given the evidence showing a rise in wage premiums for postgraduates despite a rapid expansion of advanced degree holders (Lindley & Machin, 2016), the happiness return to postgraduate qualifications may exceed that to undergraduates. Meanwhile, the psychological premium for higher education may plateau or even diminish beyond the bachelor's level. Indeed, Helliwell et al. (2012) document that postgraduates' happiness scores are only slightly higher than their undergraduate counterparts. Likewise, if highly educated individuals have stronger aspirations for better economic standing but in fact fall short of such expected levels, this prospect-reality gap may undermine their happiness (Clark et al., 2015). Note that recent research shows that postgraduate students do not always expect high economic returns and that their prospects for non-economic outcomes affect their satisfaction (Li & Jung, 2024). Nonetheless, once completing advanced programs and entering the labor market, individuals with higher degrees may encounter a problem of overqualification with their education exceeding job requirements (Erdogan & Bauer, 2021; Jung, 2023), although its risk could be relatively lower than undergraduates (Chevalier & Lindley, 2009). Considering that perceived overeducation is likely to hinder subjective well-being (Erdogan et al., 2018), we hypothesize:

Hypothesis 3: The happiness return to postgraduate degrees is smaller than, or at best equal to, undergraduate degrees. This limited premium for postgraduates is more pronounced in recent years with the rapid expansion of postgraduate programs.

Data and methods

Data and variables

We use data from the General Social Survey (GSS) in the USA over 50 years. GSS is a nationally representative survey of individuals ages 18 and over. Since its launch in 1972, GSS has been conducted annually until 1994 with some exceptions, followed by biannual implementations in even-numbered years except 2020, which was postponed to 2021. The latest available data were collected in 2022, and the current study uses all cases ages 25 to 64 (i.e., working-age samples who have generally completed the initial schooling stage) with valid information on all the variables shown in Table 1 ($N = 35,483$).

Figure 1 shows a directed acyclic graph (DAG) consisting of education (E), occupational attainment (O), income (I), happiness (H), and covariates (C). In this model, we treat O and I as the primary and secondary mediators, respectively, assuming that occupations may affect happiness both directly ($E \rightarrow O \rightarrow H$) and indirectly by enhancing income ($E \rightarrow O \rightarrow I \rightarrow H$). This framework also reflects the possibility that higher education leads to happiness via increased income without being mediated by occupations ($E \rightarrow I \rightarrow H$).

Education is quantified by a binary measure for college completion, following recent studies on education premiums (i.e., 1 is assigned to respondents who have attained a first degree) (Brand, 2023). This E measure is replaced with an advanced degree for postgraduate analysis. Occupational attainment is quantified by prestige scores ranging from 0 (bottom) to 100 (top) based on the 2010 Census Occupation Classification. Income refers to the logarithm of inflation-adjusted household income so that the effect of higher education on monetary rewards, including the one via union formation (e.g., marriage with an equally educated partner), is comprehensively estimated. This strategy also permits an analysis of individuals who are outside the labor market, thus capturing the overall contribution of college completion (Araki & Olivos, 2024; Oishi et al., 2011). Happiness is the state of feeling “very happy” on the three-point happiness scale (i.e., very happy, pretty happy, not too happy), given the response distribution (approximately 30%, 55%, and 15% for each answer option in this order), data availability over decades, and consistency with prior research (Hout, 2012). Covariates consist of age (absolute value), sex (male dummy), race/ethnicity (whites dummy), parental education (college completion dummy for a father and a mother, respectively), and statuses when respondents were 16 years old, including family income (5-point Likert scale from “far below average” to “far above average”), residential area (large city dummy), and whether respondents lived with both parents. To see the longitudinal trends whilst securing robust results, data from several years are merged in consideration of the sample size, and year-fixed effects are further incorporated. Note that changing the thresholds for year groups did not affect the main findings. See Table 1 for descriptive statistics.

Analytic strategy

To estimate causal effects, we use the counterfactual framework, an approach focused on the difference between two potential outcomes with and without specific interventions/exposures. Importantly, it is impossible to compare the same individual with two opposite

Table 1 Descriptive statistics

Variables	1973–1977	1978–1983	1984–1988	1989–1994	2002–2008	2010–2014	2016–2018	2021–2022
Happiness								
Very happy	35.8%	31.1%	30.6%	31.1%	29.9%	29.0%	29.0%	20.4%
Pretty happy	52.6%	56.9%	57.8%	57.9%	57.2%	56.8%	56.5%	56.7%
Not too happy	11.6%	12.0%	11.6%	11.0%	13.0%	14.2%	14.5%	22.9%
Educational attainment								
Below bachelor	83.1%	81.6%	77.5%	74.0%	69.4%	68.1%	67.0%	56.5%
Bachelor	11.3%	12.4%	15.4%	18.0%	19.9%	20.3%	21.4%	26.1%
Postgraduate	5.6%	6.0%	7.1%	8.0%	10.7%	11.6%	11.6%	17.4%
Occupational prestige (0–100)	42.68 (12.83)	42.87 (12.57)	43.61 (12.92)	43.94 (12.64)	44.30 (12.95)	44.18 (13.50)	44.72 (13.48)	46.23 (13.70)
Household income (log)	10.51 (0.76)	10.45 (0.85)	10.44 (0.88)	10.50 (0.87)	10.58 (0.97)	10.40 (1.11)	10.44 (1.09)	10.54 (1.16)
Age	42.40 (11.64)	41.19 (11.73)	41.06 (11.34)	41.24 (10.67)	43.22 (10.91)	43.97 (11.40)	44.28 (11.56)	44.85 (11.53)
Sex								
Male	45.6%	44.5%	45.0%	44.9%	47.8%	45.7%	45.1%	44.8%
Female	54.4%	55.5%	55.0%	55.1%	52.2%	54.3%	54.9%	55.2%
Race/ethnicity								
White	88.9%	84.5%	81.1%	84.0%	75.7%	73.9%	71.3%	74.9%
Non-white	11.1%	15.5%	18.9%	16.0%	24.3%	26.1%	28.7%	25.1%
Tertiary educated father	28.5%	33.7%	33.9%	35.9%	39.3%	42.6%	45.2%	47.5%
Tertiary educated mother	18.6%	19.7%	18.5%	18.6%	23.4%	24.6%	27.4%	31.5%
At age 16								
Family income	2.75 (0.83)	2.76 (0.85)	2.80 (0.86)	2.81 (0.86)	2.76 (0.92)	2.77 (0.93)	2.73 (0.97)	2.75 (0.97)
Lived in a large city	16.4%	15.5%	15.4%	15.0%	15.3%	15.9%	16.6%	17.9%
Lived with both parents	77.2%	74.8%	75.5%	74.5%	69.1%	64.5%	61.0%	66.1%
N	4876	4151	5003	4900	4763	4289	3332	4169

Note: The means with standard deviations (for continuous variables) and the percentages of corresponding categories (for categorical variables) are displayed. Occupational prestige is a continuous measure ranging from 0 (bottom) to 100 (top) based on the 2010 Census Occupation Classification. Household income is the logarithm of inflation-adjusted family income in constant US dollars. Family income at age 16 is a 5-point subjective status from “far below average” to “far above average”

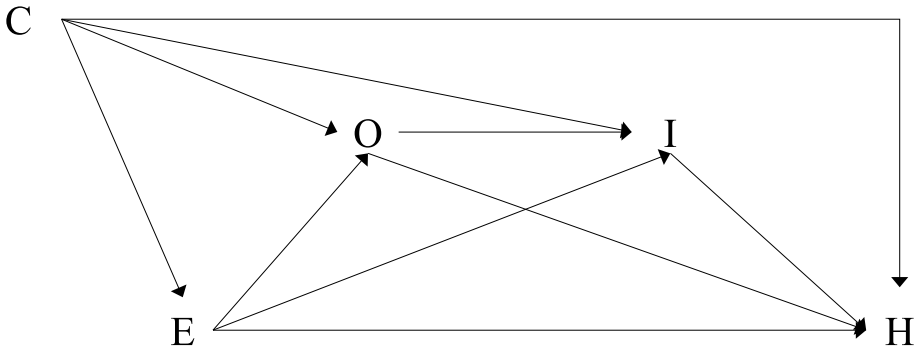


Fig. 1 Directed acyclic graph. Note: E, O, I, H, and C denote education, occupational attainment, income, happiness, and covariates

scenarios (e.g., if a person has a tertiary degree, we cannot observe their economic status and happiness level without a tertiary degree). A typical strategy here is to identify and estimate the average effect by leveraging the target population and contrasting the outcome that is naturally expected if an individual receives an intervention/exposure (e.g., tertiary educated) and the one that would have taken without it (e.g., not tertiary educated) (i.e., natural effects). In the current case, the total effect of college completion on happiness can be defined as the mean difference in the likelihood of feeling very happy between college graduates (or postgraduates), $Y(1)$, and their less-educated counterparts, $Y(0)$: $\mathbb{E}\{Y(1) - Y(0)\}$. As discussed, it is logical to consider that occupations and income operate as the primary and secondary mediators in connecting higher education and happiness.

Let $Y(e, O(ef), I(ef, O(ef)))$ denote the counterfactual outcome (happiness) that would have been should respondents' education be e and occupation/income set to the natural value they would have taken had respondents' education been e' . The overall happiness return can be decomposed into the direct effect and joint mediator $\{O, I\}$ paths as follows.

$$\begin{aligned}\mathbb{E}[Y(1) - Y(0)] &= \mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(0, O(0), I(0, O(0)))] \\ &= \mathbb{E}[Y(1, O(0), I(0, O(0))) - Y(0, O(0), I(0, O(0)))] \\ &\quad + \mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(1, O(0), I(0, O(0)))]\end{aligned}$$

where the expression $(\mathbb{E}[Y(1, O(0), I(0, O(0))) - Y(0, O(0), I(0, O(0)))])$ represents the natural direct effect ($E \rightarrow H$) and the expression $(\mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(1, O(0), I(0, O(0)))])$ indicates the joint natural indirect effect including transmissions along either one or both mediators ($E \rightarrow O \rightarrow H$; $E \rightarrow I \rightarrow H$; $E \rightarrow O \rightarrow I \rightarrow H$). The indirect path here needs to be further decomposed into the occupation effect (i.e., the composite of $E \rightarrow O \rightarrow H$ and $E \rightarrow O \rightarrow I \rightarrow H$) and the income effect net of occupations ($E \rightarrow I \rightarrow H$). As argued, considering that occupations may affect happiness not only directly but also through income ($E \rightarrow O \rightarrow I \rightarrow H$), this needs to be treated as a part of the indirect effect via occupational attainment lest we underestimate its mediating role (see Zhou & Yamamoto, 2023 for similar arguments on how to treat the indirect path through primary and secondary mediators). The indirect effects are therefore defined as follows.

$$\begin{aligned}\mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(1, O(0), I(0, O(0)))] &= \mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(1, O(0), I(1, O(0)))] \\ &\quad + \mathbb{E}[Y(1, O(0), I(1, O(0))) - Y(1, O(0), I(0, O(0)))]\end{aligned}$$

where the expression $(\mathbb{E}[Y(1, O(1), I(1, O(1))) - Y(1, O(0), I(1, O(0))])$ shows the paths along occupations, solely or jointly through income, and the expression $(\mathbb{E}[Y(1, O(0), I(1, O(0))) - Y(1, O(0), I(0, O(0))])$ is the natural indirect effect via income net of occupations.

To identify these causal effects, we need to take into account the following assumptions for no unmeasured confounding of the link between H and (E, O, I) , that between E and O , and that between (E, O) and I , as well as no H -(O, I) confounder induced by E . C represents baseline covariates.

$$Y(e, o, i) \prod E|C$$

$$Y(e, o, i) \prod O|E, C$$

$$Y(e, o, i) \prod I|E, O, C$$

$$O(e) \prod E|C$$

$$I(e, o) \prod O|E, C$$

$$I(e, o) \prod E|C$$

Were these conditions met, the causal effects can be estimated as follows. To decompose the indirect effects, following Steen et al. (2017) that proposed a flexible approach without sensitivity parameters or parametric assumptions, we first prepare extended datasets by replicating the observed data four times, with three auxiliary variables for observed and counterfactual respondents' education (e , e' , and e'') being assigned to each replication. We then compute weights and imputed nested counterfactuals $Y_i(e, O_i(e'), I_i(e'', O_i(e')))$ as fitted values $\hat{\mathbb{E}}(Y_i|E = e, O_i, I_i, C_i)$ from the outcome model below. Regressing the imputed outcome with the weights, we fitted the natural effects model for $\mathbb{E}\{Y(e, O_i(e'), I_i(e'', O_i(e')))|C\}$. Standard errors are estimated by the bootstrap method (1000 replications).

$$P(Y = 1|E, O, I, C) = \mathcal{O}(\beta_0 + \beta_1 E + \beta_2 O + \beta_3 I + \beta_4 C)$$

$$f(O|E, C) = N(\theta_0 + \theta_1 E + \theta_2 C, \sigma^2)$$

$$f(I|E, O, C) = N(\gamma_0 + \gamma_1 E + \gamma_2 O + \gamma_3 C, \sigma^2)$$

Results

Table 2 summarizes the direct and indirect effects, while Figs. 2, 3, and 4 illustrate the trend of total effects, their decomposition, and proportions of indirect effects via occupation and income, respectively. As depicted in Fig. 2 (squares

with the solid line), the overall happiness return to college completion has approximately quadrupled over 45 years since the mid-1970s. The total effect estimates rose from 0.022 in 1973–1977 to 0.090 in 2016–2018. This means that (1) university graduates are more likely than their less educated counterparts to feel very happy in general and (2) the difference in probability of reporting happiness between the two education groups increased from 2.2% to 9.0%. However, this upward trend ended in the wake of the COVID-19 pandemic, with the total effect plunging to 0.030 in 2021–2022.

The results of decomposition analysis (Table 2, Panel A, and Figs. 3–4) show nuanced trends. In 1973–1977, the indirect pathway through occupational attainment is the primary mechanism connecting higher education and happiness, with the indirect effect estimate being 0.045 (95% CI 0.021–0.067). In contrast, the contribution of the path mediated solely by income is 0.014 (95% CI 0.009–0.020), and the direct effect estimate even turns out to be negative: -0.037 (95% CI -0.080 – 0.008). This suggests that (1) the positive effect of higher education on happiness is largely driven by economic benefits; and thus, (2) college completers may feel less happy in case higher education does not bring greater economic returns, such as a better job and higher income. Nevertheless, the indirect effect through occupations fluctuates since then, while the direct pathway and the mediating effect of income net of occupations have become more pronounced except for the latest period (and 2010–2014 for the direct effect). In 2016–2018, for example, the effect estimates for direct and indirect paths are 0.033, 0.016 (via occupation), and 0.041 (via income), respectively, although the indirect effect through occupations remains statistically significant at the 10% level (see Fig. 3, Panel A, 2016–2018). Figure 4 (Panel A) clearly displays the contrasting tendency, with the decreasing and increasing mediating roles of occupations and income, the latter of which explains approximately 75% of the overall indirect effects in recent years. Hypotheses 1 and 2–2 are thus supported, whereas Hypothesis 2–1 is not confirmed except for the latest period under the influence of the pandemic.

Table 2 (Panel B) and Figs. 3–4 (Panel B) illustrate the analysis results for postgraduate degrees. The total happiness return to postgraduates has aligned with that to undergraduates from the 1970s to the 2000s. However, it has stalled since then and fluctuated until recently (see Fig. 2, triangles with the dashed lines). This supports Hypothesis 3 in that the psychological premium for advanced programs, unlike first degrees, has not been promoted over the past two decades. This decoupling lies in the limited direct effect of postgraduate degrees (e.g., 0.010, 95% CI -0.040 – 0.062 , in 2016–2018) and the indirect effect through income (e.g., 0.023, 95% CI 0.016–0.031, in 2016–2018). Meanwhile, the mediating role of occupations remains relatively high (e.g., 0.034, 95% CI 0.013 – 0.056, in 2016–2018), although it does not show an upward trend either. Consequently, the indirect pathway along occupational attainment continuously accounts for the majority of the overall indirect effect of postgraduate degrees (see Fig. 4, Panel B). This means that, for postgraduate degree holders, the transition to more prestigious occupations plays a substantial role in gaining happiness compared to tertiary graduates as a whole.

Table 2 Direct and indirect effects of higher education on happiness

	1973–1977	1978–1983	1984–1988	1989–1994	2002–2008	2010–2014	2016–2018	2021–2022
Panel A: Bachelor and above								
Direct	– 0.037 (– 0.08–0.01)	0.011 (– 0.03–0.05)	0.003 (– 0.03–0.04)	0.016 (– 0.02–0.05)	0.024 (– 0.01–0.06)	– 0.001 (– 0.04–0.03)	0.033 (– 0.01–0.07)	– 0.011 (– 0.04–0.02)
Indirect via occupation	0.045 (0.02–0.07)	0.016 (– 0.01–0.04)	0.024 (0.01–0.04)	0.025 (0.01–0.04)	0.022 (0.00–0.04)	0.039 (0.02–0.06)	0.016 (– 0.01–0.04)	0.013 (0.00–0.03)
Indirect via income (net of occupation)	0.014 (0.01–0.02)	0.017 (0.01–0.02)	0.022 (0.02–0.03)	0.023 (0.02–0.03)	0.031 (0.02–0.04)	0.032 (0.02–0.04)	0.041 (0.03–0.05)	0.028 (0.02–0.04)
Panel B: Postgraduate								
Direct	– 0.016 (– 0.08–0.05)	– 0.006 (– 0.07–0.06)	0.013 (– 0.04–0.07)	0.013 (– 0.04–0.06)	0.014 (– 0.03–0.06)	– 0.041 (– 0.09–0.00)	0.010 (– 0.04–0.06)	0.003 (– 0.03–0.04)
Indirect via occupation	0.045 (0.02–0.07)	0.022 (0.00–0.05)	0.032 (0.01–0.05)	0.040 (0.02–0.06)	0.039 (0.02–0.06)	0.060 (0.04–0.08)	0.034 (0.01–0.06)	0.019 (0.00–0.03)
Indirect via income (net of occupation)	0.011 (0.00–0.02)	0.015 (0.01–0.02)	0.015 (0.01–0.02)	0.018 (0.01–0.02)	0.021 (0.02–0.03)	0.018 (0.01–0.02)	0.023 (0.02–0.03)	0.012 (0.01–0.02)

Note: 95% confidence intervals are in parentheses, estimated using the bootstrap method (1000 replications). See Table 1 for the number of observations

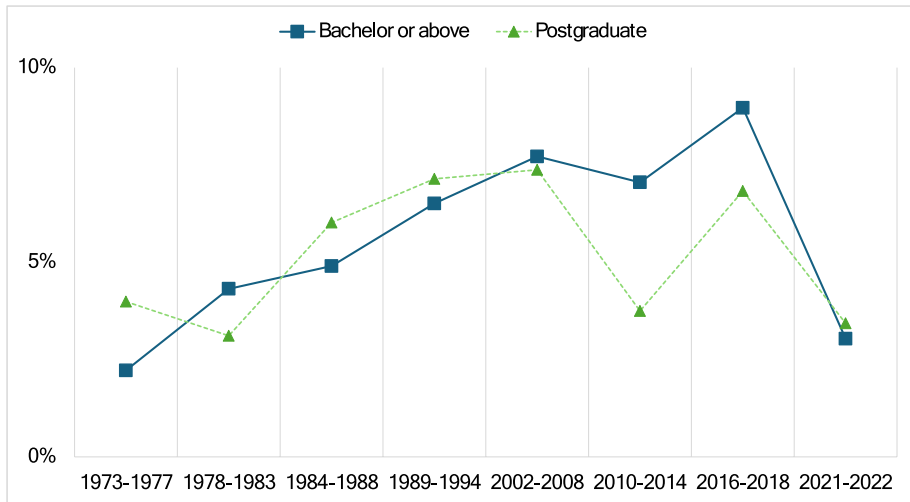


Fig. 2 Shifting happiness returns to higher education (total effects). Note: The solid and dashed lines indicate total effects of bachelor or above (squares) and postgraduate (triangles) on happiness, respectively, in terms of the difference in probability of feeling “very happy” between individuals with each educational attainment and those with lower educational attainment

Discussion and conclusion

While a growing body of research pays attention to public good outcomes of higher education, evidence remains elusive concerning the causal effect of college completion on happiness, an essential form of non-pecuniary public good (Mihut, 2024). In this article, with causal mediation analysis using the GSS dataset in the USA over 50 years, we find that tertiary graduates are more likely than their less educated counterparts to feel very happy. Importantly, this happiness return to higher education has increased over time, despite the turmoil in recent years under the COVID-19 pandemic. Looking at decomposition, in the mid-1970s, the psychological premium for higher education is primarily attributed to occupational attainment, whereas the effect estimate net of economic benefits shows a negative sign. Nonetheless, the relative impact of the indirect path through occupations has gradually diminished in contrast to the increasing mediating role of income per se over the years. Meanwhile, the happiness return to postgraduate degrees, which has been realized primarily by the indirect pathway through occupations, has stalled since the 2000s.

These findings indicate several key implications. First, higher education may contribute to relatively better non-pecuniary outcomes like subjective well-being, although its magnitude fluctuates. From a policy perspective, promoting higher education opportunities therefore potentially enhances public good outcomes, at least in terms of human development as assessed by happiness levels. Nevertheless, it is crucial to note that the happiness premium for college completion somewhat reflects the marginalization of less-educated individuals, whose chance of securing happiness is on the decline over the past decades. Figure 5 indeed shows that the percentage of the less educated reporting “very happy” has been decreasing while their tertiary-educated counterparts exhibit a stable trend, resulting in widening gaps between them, except during the pandemic period. This indicates that, as widely observed in the exacerbated economic disadvantage among less-educated people amidst higher education expansion (Boudon, 1974),

[A Bachelor or above]

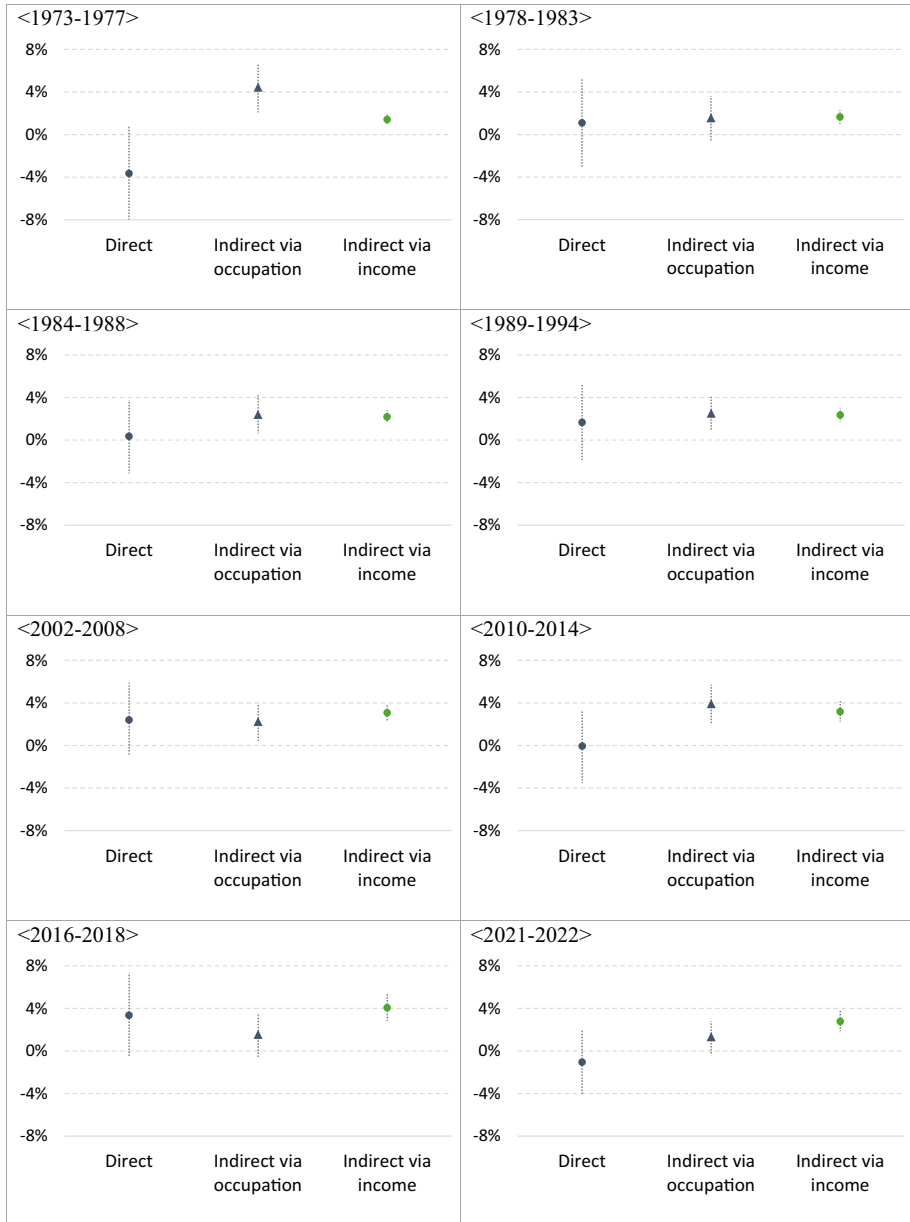


Fig. 3 Direct and indirect effects of higher education on happiness. Panel A: Bachelor or above. Panel B: Postgraduate. Note: The dots represent the direct and indirect effects via occupation and income (net of occupation) in terms of the difference in probability of feeling “very happy” between individuals with each educational attainment and those with lower educational attainment, with the bars indicating the 95% confidence intervals, estimated by the bootstrap method (1000 replications)

[B Postgraduate]

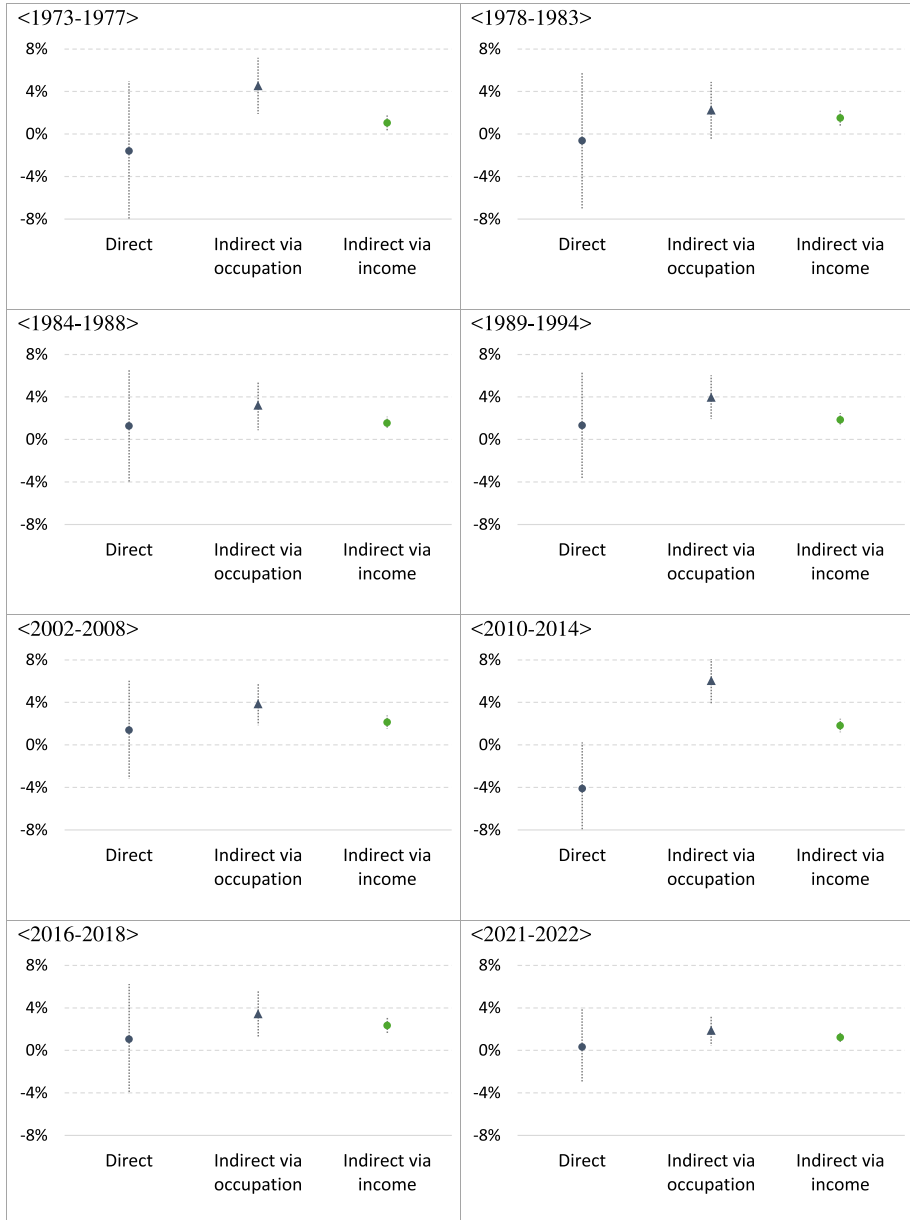
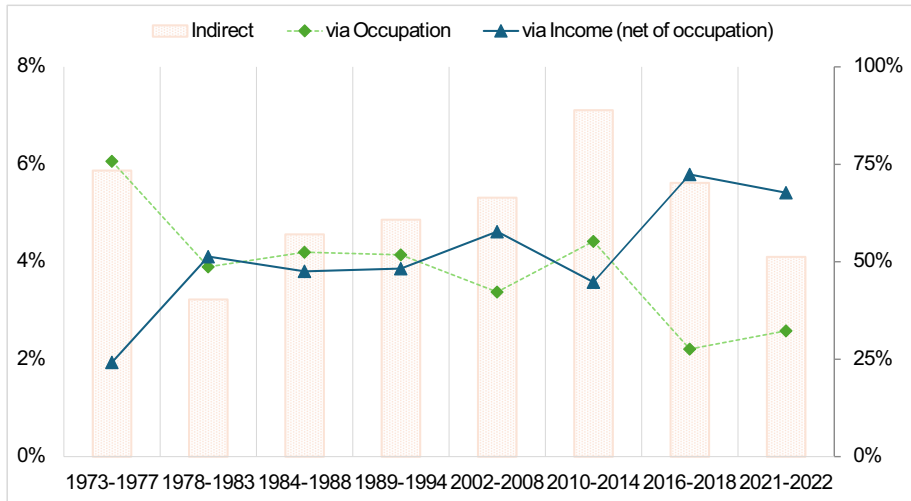


Fig. 3 (continued)

they may have also been penalized psychologically, and the education-based inequalities have intensified multidimensionally. These tendencies highlight the need to carefully consider the potential trade-off between positive and adverse effects of fostering higher education on college completers and non-completers lest increased opportunities lead to their socio-economic and psychological disparities.

[A Bachelor or above]



[B Postgraduate]

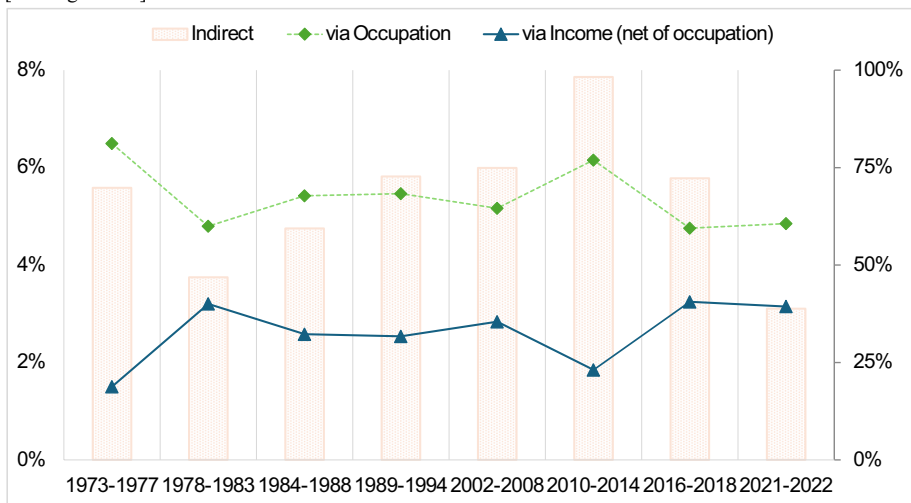


Fig. 4 Proportions of indirect effects via occupation and income. Panel A: Bachelor or above. Panel B: Postgraduate. Note: The bars show the indirect effect of higher education on happiness in terms of the difference in probability of feeling “very happy” between individuals with each educational attainment and those with lower educational attainment (left Y axis), including both paths via occupation and income. The dashed and solid lines illustrate the proportions of indirect effects via occupation and income, respectively (right Y axis)

Second, economic benefits play an essential role in forming the happiness return to higher education. For a bachelor or above, the mediating power of income net of jobs is prevalent, whereas the indirect path through occupations is key for postgraduate degrees. This means that university graduates in contemporary America have a certain chance of gaining monetary rewards bypassing occupations, resulting in a relatively higher probability of feeling happy. Meanwhile, the same mechanism rarely operates for advanced degree

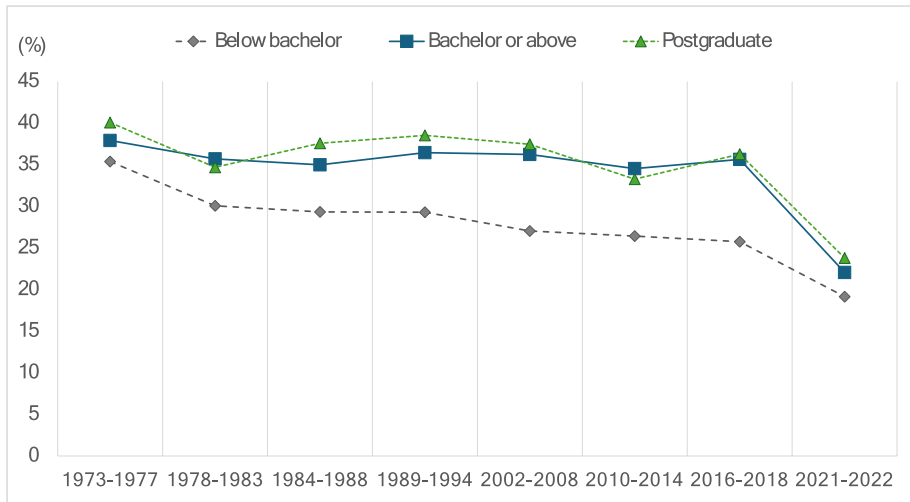


Fig. 5 Percentage of feeling very happy across education groups. Note: The solid, dotted, and dashed lines with rectangles, triangles, and diamonds indicate the percentages of respondents feeling “very happy” among those with a first degree (bachelor or above), an advanced degree, and neither of them, respectively. The values are simply calculated by dividing the number of “very happy” people by the sample size for each group using the same dataset for causal mediation analysis

holders, whose happiness largely depends on their occupational attainment. Although further analyses are necessary to explain this structure, one may assume that postgraduates are more likely than undergraduates to be rewarded by certain jobs requiring specific and advanced knowledge and skills cultivated by specialized programs, and such occupations may enhance both income and satisfaction. At the same time, should the occupational prestige as well as pecuniary returns be relatively higher among postgraduates, the attained social status coupled with better job quality may contribute to their subjective well-being both symbolically and instrumentally. In either case, the overall happiness premium for higher education, both at the undergraduate and postgraduate levels, may possibly vanish once their economic rewards become less pronounced.

From a comparative viewpoint, it is noteworthy that the said findings could be unique to the USA. For example, while the current study finds an upward trend of mediation by labor market outcomes enhanced by higher education, recent comparative studies show that the economic values of credentials tend to decline in the context of educational proliferation and skills diffusion (Araki, 2020; Araki & Kariya, 2022). Likewise, the increasing tendency that postgraduate degree holders encounter difficulty in obtaining proper jobs has been widely reported in non-American societies (Jung, 2023). Should this be the case, one may rather observe the stalled, or even declining, happiness return to higher education in other societies. Notwithstanding, it is still possible that the psychological premium for higher education exists in such circumstances so long as the “direct” effect is more prevalent than in the USA. This includes both (1) the genuinely direct contribution of higher education to happiness, in that high credentials as such and/or the learning process per se, coupled with interactions with peers and teachers, provide tertiary graduates psychological benefits, and (2) the indirect pathways that are not included in the current empirical model but operate as key mechanisms connecting higher education and happiness. The latter encompasses health-related behaviors and outcomes, social ties strengthened

through advanced educational experiences, and family formation (e.g., marriage fostered by educational assortative mating). In particular, the mediating role of health is worthy of investigation, given that abundant evidence shows (1) health determines happiness (Helliwell et al., 2024) and (2) the higher level of education leads to better health conditions through greater access to healthcare services, less engagement in unhealthy behaviors, and more social support to cope with stress (C. Ross & Wu, 1995; von dem Knesebeck et al., 2006).

To advance this line of scholarship and provide more implications for relevant policy and practice, future research should incorporate diverse variables for higher education. Indeed, prior studies consistently find that the value of tertiary degrees and the risk of over-education significantly varies across higher education institutions (including their prestige) and fields of study (Barone & Ortiz, 2011; Britton et al., 2020; Horowitz & Ramaj, 2024; Iannelli et al., 2018; Kim et al., 2015; Marginson, 2016c; Ochsenfeld, 2014; Sullivan et al., 2018; Van de Werfhorst & Luijkx, 2010). It is therefore likely that the happiness return to higher education differs when taking account of its vertical and horizontal differences. Likewise, happiness indicators can be extended beyond the three-point scale used in our analysis, including the typical 0–10 Cantril Ladder scale for life satisfaction and frequency measures of positive and negative affect.

Furthermore, given the recent plunge during the pandemic, longer time periods should be analyzed to examine if the happiness return rebounds or remains stalled in post-pandemic times, preferably using panel data. This is important to establish more robust causal mechanisms as the dataset used in our analysis is based on repeated cross-sectional surveys and does not fully account for time-invariant individuals' attributes/traits. Put differently, unobserved personal factors of selecting into tertiary programs need to be taken into account to better explain the causal path from higher education to happiness. In so doing, as recent studies demonstrate (Brand & Xie, 2010; Xie et al., 2012; Zhou & Xie, 2020), one should pay close attention to effect heterogeneity across individuals with different backgrounds (e.g., gender, race/ethnicity, and parental status), exploring who is more likely to gain psychological benefits from higher education and whether social inequality is exacerbated or mitigated due to the heterogeneous effects. As an overarching perspective, the influence of education policy, labor market structure, and socio-cultural conditions on these dynamics should be investigated through comparative studies. With these future directions, our framework and empirical findings lay the foundation to better understand private and public good outcomes of higher education.

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Data availability Data are available from the following website: General Social Survey (<https://gss.norc.org/>).

Code availability Codes are available from the authors upon request.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Conflicts of interest The authors declare no competing interests.

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